

Small Signal Stability Analysis with High Penetration of Grid-connected Wind Farm of PMSG Type Considering the Wake Effect

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Abstract— This paper aims to discuss the impacts of high penetration of grid-connected wind farm of permanent magnet synchronous generator (PMSG) type on the small signal stability of power systems with wake effect taken into account. Firstly, the modeling of PMSG as well as its converters is described. The modeling of wind farm wake effect and the corresponding wind farm equivalent model are given in detail. Then, the process of deducing the system state matrix is introduced. Finally, the simulations with multiple scenarios consideration are carried out on the IEEE 3-generator-9-bus test system as benchmark to reveal the impacts of wake effect on small signal stability.

Index Terms-- PMSG; wake effect; state matrix; small signal stability analysis

I. INTRODUCTION

Energy shortage and environmental pollution are worldwide issues. The advent of renewable clean energy would be a potential alternative for the solution of the issues. Especially, in recent years the wind power has gradually become the most prominent renewable power source. The total world installed wind power capacity has reached 282GW in 2012[1]. And the installed capacity in China is 75324.2MW in 2012[2]. Accordingly, the fluctuation and randomness of wind generation caused by the intermittent wind speed may lead to a great influence on the power system stability problems like small signal stability with high penetration of wind power. More and more scholars and electrical engineers have gaining intensive attention to this issue. Take simplified Norway power grid as an example, the inter area power oscillation modes pertinent to different types of wind power generators were analyzed [3]. In the study of the small signal stability analysis with high penetration of grid-connected wind farm of doubly-fed induction generator (DFIG) type, the wind turbine generator (WTG) can provide a good damping characteristic, but there are exceptions [4-6]. A mathematical model of asynchronous wind power generator for small signal stability analysis was proposed and can be found in [7]. Reference [8] discussed the small signal stability of grid-connected PMSG wind farm with or without PI control. So far, achievements of small signal stability analysis

mainly focus on the DFIG type. And in general, the wind farm is equivalent to a macro generator during analysis. In fact, owing to the wake effect [9], the wind speed captured by each WTG in the wind farm is different. Hence, it is imperative to further conduct the research of small signal stability of power system incorporating wind power with wake effect taken into account.

In this paper, we aim to reveal the influence of grid-connected wind farm of PMSG type on power system small signal stability. The modeling of PMSG and wind farm with wake effect is given. The IEEE 3-generator-9-bus test system is employed as benchmark to carry out the small signal stability analysis, and the comparisons with and without wake effect consideration are given as well.

II. MODELING OF PMSG AND WIND FARM

A. Modeling of wind turbine

The mechanical power extracted from the wind can be expressed as:

$$P_m = \begin{cases} 0 & v_{\omega 0} \leq v_{cut-in} \text{ or } v_{\omega 0} \geq v_{cut-off} \\ \frac{1}{2} \rho \pi R^2 C_p(\beta, \lambda) v_{\omega 0}^3 & v_{cut-in} < v_{\omega 0} \leq v_{rated} \\ P_r & v_{rated} < v_{\omega 0} \leq v_{cut-off} \end{cases} \quad (1)$$

where, P_m is the power extracted from wind; ρ is the air density; $v_{\omega 0}$ is the wind speed; R is the radius of the wind turbine; C_p is the power factor; β is the blade pitch angle; λ is the tip speed ratio; v_{cut-in} is the cut-in speed; $v_{cut-off}$ is the cut-off speed; v_{rated} is the rated wind speed.

B. Modeling of PMSG

The PMSG model is established according to the generator convention. The rotor flux orientated control strategy is adopted during modeling.

The electromagnetic power generated by the PMSG is as follows:

$$\begin{cases} P_s = u_{ds} i_{ds} + u_{qs} i_{qs} \\ Q_s = u_{qs} i_{ds} - u_{ds} i_{qs} \end{cases} \quad (2)$$

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where u_{ds} , u_{qs} are the stator voltages of d and q axis respectively.

The electromagnetic torque is as follows:

$$T_e = \frac{P_s}{\omega_s} = \frac{1}{\omega_s} (u_{ds} i_{ds} + u_{qs} i_{qs}) \quad (3)$$

It is assumed that the transient process of stator flux can be ignored during small signal stability analysis. Furthermore, one mass shaft model is used to simplify the rotor motion equations:

$$\begin{cases} 2H \frac{d\omega_e}{dt} + D(\omega_e - 1) = T_m - T_e \\ \frac{d\delta}{dt} = \omega_B(\omega_e - 1) \end{cases} \quad (4)$$

where D is the damping coefficient; H is the inertia constant of the whole drive train; ω_B is the reference value of the angular speed; T_m is the mechanical torque of generator.

C. Modeling of converters

The converter model of PMSG mainly includes 3 parts: generator-side converter, the DC part and the grid-side converter. The model is shown in Fig.1

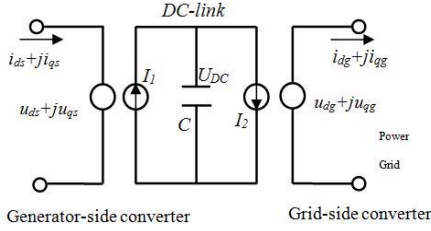


Fig.1 Model of converter and control system

Ignoring the power losses of the generator-side and the grid-side converters, the model of DC link can be derived as

$$CU_{DC} \frac{dU_{DC}}{dt} = (u_{ds} i_{ds} + u_{qs} i_{qs}) - (u_{dg} i_{dg} + u_{qg} i_{qg}) \quad (5)$$

The block diagram of the generator-side is shown in Fig.2

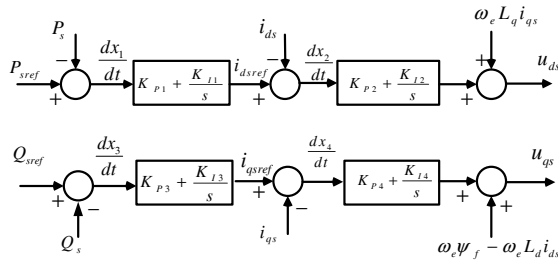


Fig.2 Generator-side control system

Where x_1 , x_2 , x_3 , x_4 are the intermediate variables; K_{P1} , K_{P2} , K_{P3} , K_{P4} are amplification factors; K_{I1} , K_{I2} , K_{I3} , K_{I4} are integral factors; i_{dsref} , i_{qsref} are the specified d-axis and q-axis stator current reference values; P_{sref} , Q_{sref} are the active power and reactive power control reference respectively. The corresponding equations are as follows:

$$\frac{dx_1}{dt} = P_{sref} - P_s \quad (6)$$

$$i_{dsref} = K_{P1}(P_{sref} - P_s) + K_{I1}x_1 \quad (7)$$

$$\frac{dx_2}{dt} = i_{dsref} - i_{ds} = K_{P1}(P_{sref} - P_s) + K_{I1}x_1 - i_{ds} \quad (8)$$

$$u_{ds} = K_{P2}(K_{P1}(P_{sref} - P_s) + K_{I1}x_1 - i_{ds}) + K_{I2}x_2 + \omega_e L_q i_{qs} \quad (9)$$

$$\frac{dx_3}{dt} = Q_{sref} - Q_s \quad (10)$$

$$i_{qsref} = K_{P3}(Q_{sref} - Q_s) + K_{I3}x_3 \quad (11)$$

$$\frac{dx_4}{dt} = i_{qsref} - i_{qs} = K_{P3}(Q_{sref} - Q_s) + K_{I3}x_3 - i_{qs} \quad (12)$$

$$u_{qs} = K_{P4}(K_{P3}(Q_{sref} - Q_s) + K_{I3}x_3 - i_{qs}) + K_{I4}x_4 + \omega_e \psi_f - \omega_e L_d i_{ds} \quad (13)$$

The control objective of the grid side converter is to regulate the DC link voltage and the output reactive power to the grid. The block diagram of the grid-side is shown in Fig.3

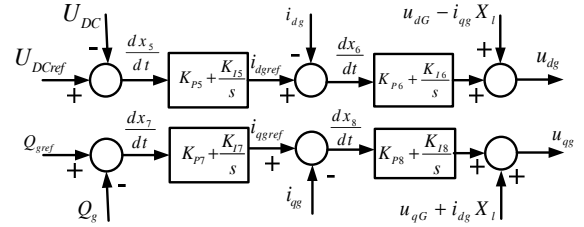


Fig.3 Grid-side control system

where x_5 , x_6 , x_7 , x_8 are the intermediate variables; K_{P5} , K_{P6} , K_{P7} , K_{P8} are amplification factors; K_{I5} , K_{I6} , K_{I7} , K_{I8} are integral factors; i_{dref} , i_{qref} are the specified d-axis and q-axis grid current reference values; U_{DCref} , Q_{gref} are the reference values of the terminal voltage of the DC link and reactive power control reference of the grid. The corresponding equations are as follows:

$$\frac{dx_5}{dt} = U_{DCref} - U_{DC} \quad (14)$$

$$i_{dref} = K_{P5}(U_{DCref} - U_{DC}) + K_{I5}x_5 \quad (15)$$

$$\frac{dx_6}{dt} = i_{dref} - i_{dg} = K_{P5}(U_{DCref} - U_{DC}) + K_{I5}x_5 - i_{dg} \quad (16)$$

$$u_{dg} = K_{P6}(K_{P5}(U_{DCref} - U_{DC}) + K_{I5}x_5 - i_{dg}) + K_{I6}x_6 + u_{dG} - i_{qg} X_l \quad (17)$$

$$\frac{dx_7}{dt} = Q_{gref} - Q_g \quad (18)$$

$$i_{qref} = K_{P7}(Q_{gref} - Q_g) + K_{I7}x_7 \quad (19)$$

$$\frac{dx_8}{dt} = i_{qref} - i_{qg} = K_{P7}(Q_{gref} - Q_g) + K_{I7}x_7 - i_{qg} \quad (20)$$

$$u_{qg} = K_{P8}(K_{P7}(Q_{gref} - Q_g) + K_{I7}x_7 - i_{qg}) + K_{I8}x_8 + u_{qG} + i_{dg} X_l \quad (21)$$

D. Modeling of wind farm

In general, a wind farm is equivalent to a macro generator for different applications. However, owing to the wake effect, the overall power output of downwind wind turbine will decrease. In this paper, the Jensen model [10] as shown in

Fig. 4 is employed to describe the wake effect in a wind farm located on flat terrain.

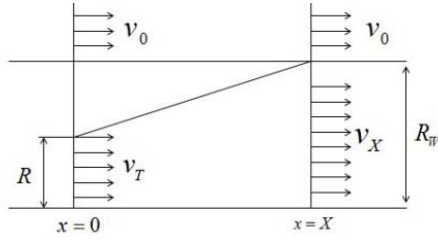


Fig.4 Jensen model

In Fig.5, X is the distance between the front row and the rear row; R_w and R are wake radius and radius of the impeller, respectively; v_0 is the initial wind speed; v_T is the wind speed that pass through the wind machine; v_X is the wind speed influenced by wake effect.

According to Jensen model, the equations are as follows:

$$v_x = v_0 [1 - (1 - (1 - C_T)^{\frac{1}{2}}) (\frac{R}{R + kX})^2] \quad (22)$$

where k is the wake loss factor; C_T is the thrust coefficient.

Taking a large-scale wind farm located on flat terrain as shown in Fig.5 for example, it is assumed that there are Y rows in the wind direction and each row has X wind turbines.

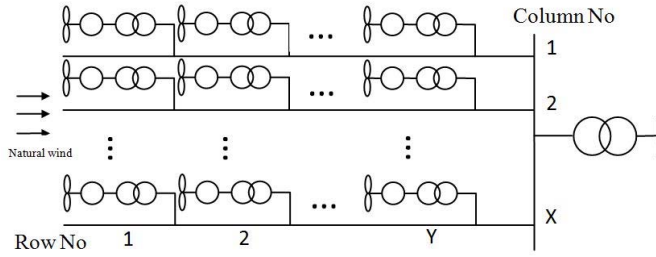


Fig.5 Topology of a wind farm

These wind turbines in the same wind speed are considered in similar operating conditions. In the simplified calculation, these wind turbines in the same row at the wind direction are considered in the same wind speed. And each row can be equivalent to one wind turbine. Fig.6 shows the layout of equivalent wind farm consisting of Y equivalent wind turbines.

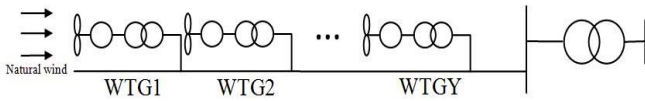


Fig.6 Equivalent wind farm based on Jensen model

III. SOLUTION OF SMALL SIGNAL STABILITY ANALYSIS INCORPORATING WIND FARM OF PMSG TYPE

For a power system, it can be written in the form of differential equations and algebraic equations as given in following:

$$\begin{aligned} \frac{dx}{dt} &= f(x, y) \\ 0 &= g(x, y) \end{aligned} \quad (23)$$

Making linearization in the vicinity of the steady state value, the equations can be written as:

$$\begin{bmatrix} \frac{d\Delta x}{dt} \\ 0 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \quad (24)$$

The system state matrix Λ can be written as:

$$\Lambda = A - BD^{-1}C$$

where A , B , C and D are Jacobian matrices of the whole system.

Assume that the system has S state vectors, N buses and the wind farm is equivalent to p wind turbines. The corresponding Jacobian matrices A , B , C and D can be expressed as:

$$\begin{aligned} A &= A_G = \begin{bmatrix} A_{G1} & & & & \\ & \dots & & & \\ & & A_{Gn} & & \\ & & & A_{WF1} & \\ & & & & \dots \\ & & & & & A_{WFp} \end{bmatrix}_{S \times S} \\ B &= \begin{bmatrix} B_{G1} & & 0 \\ & \dots & \\ & & B_{Gn} & & 0 \\ & & & B_{WF1} & \\ & & & & \dots \\ & & & & & B_{WFp} \end{bmatrix}_{S \times 2N} \\ C &= \begin{bmatrix} -C_{G1} & & & & \\ & \dots & & & \\ & & -C_{Gn} & & \\ & & & -C_{WF1} & \\ & & & & \dots \\ & & & & & -C_{WFp} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{2N \times S} \\ D &= \begin{bmatrix} Y_{GG} - D_G & Y_{GL} \\ Y_{LG} & Y_{LL} \end{bmatrix} = \begin{bmatrix} Y_{G1} - C_{G1} & Y_{G1Gn} & Y_{G1WF1} & Y_{G1WFp} & Y_{G1L} \\ M & \dots & & & \\ Y_{GnG1} & Y_{Gn} - C_{Gn} & Y_{GnWF1} & Y_{GnWFp} & Y_{GnL} \\ Y_{WF1G1} & Y_{WF1Gn} & Y_{WF1} - C_{WF1} & & Y_{WF1L} \\ M & & \dots & & \\ Y_{WFpG1} & Y_{WFpGn} & Y_{WFpWF1} & Y_{WFp} - C_{WFp} & Y_{WFpL} \\ Y_{LG1} & Y_{LGn} & Y_{LWF1} & Y_{LWFp} & Y_{LL} \end{bmatrix}_{2N \times 2N} \end{aligned}$$

where $G_1 \sim G_n$ represent the n synchronous generators; $WF_1 \sim WF_p$ represent the p equivalent wind turbines; A_{WF} , B_{WF} , C_{WF} , D_{WF} are the Jacobian matrices related to the wind farm. They can be obtained by following equations:

$$\begin{aligned} A_{WF} &= A_{PMSG} \\ B_{WF} &= B_{PMSG} Y_{WF} \\ C_{WF} &= -C_{PMSG} \\ D_{WF} &= I_{4 \times 4} - D_{PMSG} Y_{WF} \end{aligned} \quad (25)$$

IV. APPLICATION EXAMPLE

Taking IEEE-3-generator-9-bus system as example [11], two scenarios are considered during simulations: (1) the wind

farm is connected to bus 7 considering the wake effect; and (2) the wind farm is connected to bus 7 ignoring the wake effect. The parameters of PMSG used in the simulations are given as follows: Air density $\rho=1.2234\text{kg/m}^3$; Rated wind speed $v_{rated}=11.52\text{m/s}$; Cut-in wind speed $v_{cut-in}=3\text{m/s}$; Cut-off wind speed $v_{cut-off}=30\text{m/s}$; Rated line voltage $U_f=1.0\text{kV}$; Rated power $P_f=5.0\text{MW}$; Inertia constant $H=3\text{s}$; Number of pole pairs $p=75$; Frequency $f=50\text{Hz}$; d-axis reactance $X_d=1.0\text{p.u.}$; q-axis reactance $X_q=0.7\text{p.u.}$; Stator resistance $R_s=0.01\text{p.u.}$; DC capacitor $C=0.55\text{p.u.}$; Field flux $\psi_f=1.55\text{p.u.}$; Line reactance $X_f=0.4\text{p.u.}$; Wind turbine radius $R=60\text{m}$. The parameters of PI controllers used in the converters are as follows: $K_{P1}=46.1117$; $K_{P2}=200$; $K_{P3}=200$; $K_{P4}=0.0001$; $K_{P5}=3.6596$; $K_{P6}=0.0001$; $K_{P7}=0.0001$; $K_{P8}=0.0001$; $K_{I1}=200$; $K_{I2}=200$; $K_{I3}=0.0001$; $K_{I4}=24.6481$; $K_{I5}=0.0001$; $K_{I6}=0.0001$; $K_{I7}=152.8806$; $K_{I8}=0.0001$.

The system eigenvalues of with/without wake effect taken into account are given in Table I and Table II, respectively. The simulation results for the two situations show that the test system is small signal stable.

TABLE I
SYSTEM EIGENVALUES OF WIND FARM CONNECTED TO BUS 7 CONSIDERING WAKE EFFECT

λ_1	-90339	λ_{17}	-0.65-0.93i	λ_{33}	-0.0004+0.06i
λ_2	-27715	λ_{18}	-0.65+0.93i	λ_{34}	-0.0004-0.06i
λ_3	-12334	λ_{19}	-0.65-0.93i	λ_{35}	-0.56
λ_4	-51	λ_{20}	-0.65+0.93i	λ_{36}	-0.56
λ_5	-52	λ_{21}	-0.65-0.93i	λ_{37}	-0.56
λ_6	-32	λ_{22}	-0.49+0.68i	λ_{38}	-0.005
λ_7	-31	λ_{23}	-0.49-0.68i	λ_{39}	-0.004
λ_8	-0.64+12.29i	λ_{24}	-0.35+0.53i	λ_{40}	-0.004
λ_9	-0.64-12.29i	λ_{25}	-0.35-0.53i	λ_{41}	-0.000009
λ_{10}	-0.16+9.58i	λ_{26}	-0.31	λ_{42}	-0.000008
λ_{11}	-0.16-9.58i	λ_{27}	-0.26	λ_{43}	-0.000008
λ_{12}	-0.02+4.93i	λ_{28}	-0.2	λ_{44}	-0.0000002
λ_{13}	-0.02-4.93i	λ_{29}	-0.002+0.13i	λ_{45}	-0.0000002
λ_{14}	-5.97	λ_{30}	-0.002-0.13i	λ_{46}	-0.0000002
λ_{15}	-5.65	λ_{31}	-0.0005+0.06i		
λ_{16}	-0.65+0.93i	λ_{32}	-0.0005-0.06i		

TABLE II
SYSTEM EIGENVALUES OF WIND FARM CONNECTED TO BUS 7 IGNORING WAKE EFFECT

λ_1	-14103	λ_{10}	-0.01+5.21i	λ_{19}	-0.36-0.55i
λ_2	-51	λ_{11}	-0.01-5.21i	λ_{20}	-0.46
λ_3	-52	λ_{12}	-6.08	λ_{21}	-0.009+0.13i
λ_4	-32	λ_{13}	-5.66	λ_{22}	-0.009-0.13i
λ_5	-31	λ_{14}	-1.91+0.37i	λ_{23}	-0.19
λ_6	-0.838+13.01i	λ_{15}	-1.91-0.37i	λ_{24}	-0.001
λ_7	-0.838-13.01i	λ_{16}	-0.5+0.69i	λ_{25}	-0.00003
λ_8	-0.53+10.27i	λ_{17}	-0.5-0.69i	λ_{26}	-0.0000005
λ_9	-0.53-10.27i	λ_{18}	-0.36+0.55i		

From Table I and II, three EM oscillatory modes can be found in each simulation scenario. The corresponding characteristics of the EM oscillatory modes pertinent to different simulation scenarios are listed in Table III and Table IV.

TABLE III
CHARACTERISTICS OF EM FOR SCENARIO (1)

No.	EM Oscillatory	frequency(Hz)	damping	EM relative
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	mode		ratio	ratio
1	-0.644 ± 12.29i	1.956	0.052	24
2	-0.155 ± 9.585i	1.525	0.016	103
3	-0.023 ± 4.925i	0.784	0.0047	49

TABLE IV
CHARACTERISTICS OF EM FOR SCENARIO (2)

No.	EM Oscillatory mode	frequency(Hz)	damping ratio	EM relative ratio
1	-0.838 ± 13.007i	2.07	0.064	20
2	-0.53 ± 10.274i	1.635	0.051	24
3	-0.011 ± 5.208i	0.829	0.0041	131

The magnitude and phase angle of the right eigenvector of generators under each EM oscillatory mode considering the wake effect are shown in Fig.7 and Fig.8. In the right eigenvectors of the EM oscillatory mode 1, the amplitudes corresponding to the v_o of generators 1 and 2 are relatively small, whilst these two phase angles are approximate. The amplitudes corresponding to the v_o of generators 3 is larger and the phase angle has an opposite direction. Therefore, the mode 1 denotes the EM oscillation between the generator 3 and generators 1, 2. Similarly, the mode 2 denotes the EM oscillation between the generator 1 and the other two generators. The mode 1 and the mode 2 are intra-area EM oscillation. The mode 3 is the inter-area EM oscillation between the generator 1 and the other two generators.

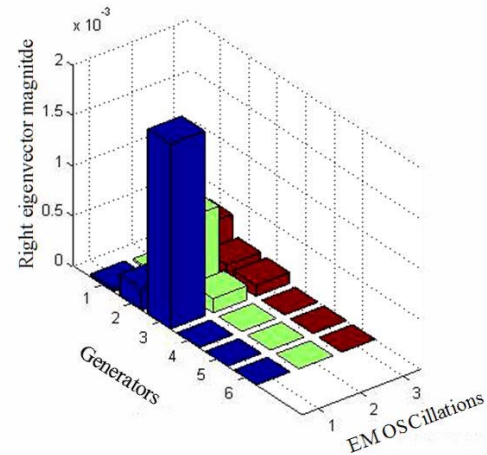


Fig.7 The right eigenvector magnitude of generators under 3 EM oscillatory modes (considering wake effect)

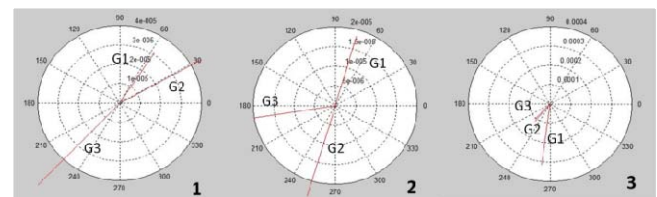


Fig.8 The right eigenvector angle of generators under 3 EM oscillatory modes (considering wake effect)

The right eigenvector magnitude and angle of generators under each EM oscillatory mode ignoring the wake effect are shown in Fig.9 and Fig.10. The mode 1 denotes the EM oscillation between the generator 1 and generators 2, 3. The mode 2 denotes the EM oscillation between the generator 3 and the other two generators. The mode 1 and mode 2 are intra-area EM oscillation. The mode 3 is the inter-area EM oscillation between the generator 1 and the other two generators.

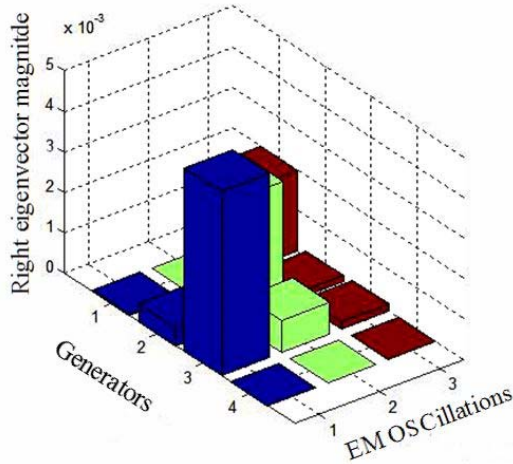


Fig.9 The right eigenvector magnitude of generators under 3 EM oscillatory modes (ignoring wake effect)

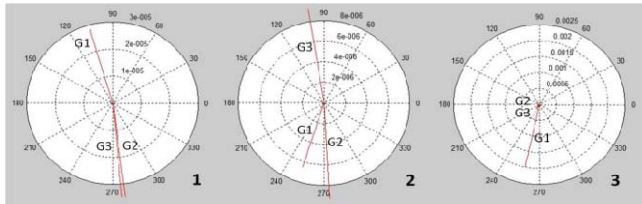


Fig.10 The right eigenvector angle of generators under 3 EM oscillatory modes (ignoring wake effect)

Table V gives the comparisons between the two simulation scenarios. Compared to the scenario considering the wake effect, the oscillation frequency mode slight increases in the case ignoring the wake effect. The damping ratio of the first two modes changed from 0.052 to 0.064 and 0.016 to 0.051. The damping ratio of the third mode remains unchanged. Comparison shows that the system damping ratio is weakened after considering the wake effect. For mode 1 and mode 2, the strong correlation generators with the oscillation in cases of considering the wake effect and ignoring the wake effect are different. Therefore, the strong correlation generators with the oscillation may change when considering the wake effect.

TABLE V
COMPARISONS BETWEEN THE TWO SCENARIOS

	MODE 1	MODE 2	MODE 3	MODE 1	MODE 2	MODE 3

	(with wake effect)	(with wake effect)	(with wake effect)	(without wake effect)	(without wake effect)	(without wake effect)
frequency(Hz)	1.956	1.525	0.784	2.07	1.635	0.892
Type of oscillation	intra-area	intra-area	inter-area	intra-area	intra-area	inter-area
damping ratio	0.052	0.016	0.0047	0.064	0.051	0.0041
Mode shape	G3 VS G1,G2	G1 VS G2,G3	G1 VS G2,G3	G1 VS G2,G3	G3 VS G1,G2	G1 VS G2,G3

V. CONCLUSION

This paper discusses the small signal stability analysis of power systems incorporating wind power considering wake effect. On the basis of PMSG model, the equivalent wind farm model with wake effect taken into account is deduced. Next, the system state matrix is derived to implement the small signal stability analysis with wake effect introduced. In order to observe the small signal stability before and after considering wake effect, the IEEE 3-generator-9-bus text system is taken as benchmark to carry out systematic simulations and comparisons. Some preliminary conclusions can be drawn that the system damping characteristics is weakened after considering the wake effect. Furthermore, the strong correlation generators with the oscillation may change when considering the wake effect.

REFERENCES

- [1] L. Qiangqiang, G.Dan, "Summary of wind power development in China," *The Electrical Age*, vol.4, pp39-40, 2013,
- [2] CWEA, "Installed capacity of wind power in China 2012," *Wind Power*, vol.3:pp44-55, 2013.
- [3] H. Espen; N. Ian; U. Kjetil, "Large scale wind power integration in Norway and impact on damping in the Nordic grid," *Wind Energy*, vol.8, pp375-384., 2005.
- [4] O. Anaya-Lara, M. Hughes, N. Jenkins, J.Strbac, "Influence of wind farms on power system dynamic and transient stability," *Wind Engineering*, vol. 30, pp. 107-27, 2006.
- [5] E.Muljadi; C.P Butterfield; B.Parsons; A.Ellis, "Effect of variable speed wind turbine generator on stability of a weak grid," *IEEE Transactions on Energy Conversion*, vol.22, no.1, pp.29-36, March. 2007
- [6] G. Tsourakis; B. M. Nomikos; C. D. Vournas, "Contribution of doubly fed wind generators to oscillation damping," *IEEE Transactions on Energy Conversion*, vol.24, no.3, pp.783-791, Sept. 2009.
- [7] J. G. Siootweg, W. L. Kling, "The impact of large scale wind power generation on power system oscillations," *Electric Power Systems Research*, vol. 67, no. 1, pp. 9-20, Oct, 2003.
- [8] H. H. H. Aly, M. E. El-Hawary, "Small signal stability analysis of tidal current turbine using ddpmsg with and without controller," *IEEE Electrical Power and Energy Conference*, 2011, pp.239-243..
- [9] S. Jianfeng, "Research of modeling and simulation of wind farm," Ph.D, dissertation, Tsinghua University. Beijing. 2004.
- [10] Z. Ruimin, L. Jianhua, L. Zuohong, L. Yihua, "The model of wind farm considering the wake effect and the calculation of the stochastic power flow," *Journal of Xi'an Jiao Tong University*, pp.1515-1520, 2008,
- [11] P. Kundur, "Power system stability and control", New York: McGraw-Hill, 1994.